prop:bialgebra-Mod-monoidal in Chapter 7

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aljabr2:e03224e2e867a49dc60c7975d890079068a30466:chapter7; Creative Commons Attribution 4.0 International (CC BY 4.0)

Editorial note. All mathematical statement and proof text is reproduced verbatim from chapter7.tex of the pinned checkout github.com/wenweili/AlJabr-2 at commit e03224e2e867a49dc60c7975d890079068a30466. The file is packaging: it adds no mathematics and rewrites no author text.

Editorial note. Packaging changes, each forced by the pinned TeX Live 2025 and each checked before writing: the author's class needs ctex, xeCJK, zhnumber and ntheorem, which are not installed, so the Chinese text is set with the Noto Serif / Noto Sans CJK SC faces that are installed – the same treatment the three pre-existing AlJabr-2 deliveries in this corpus use; the author's own printed numbers are restored from a per-task fragment; the index commands are no-ops.

Editorial note. The counted span keeps the author's cross-references to results outside it. Those results are the author's earlier results in the same pinned commit; they are pointed at and not reproduced, and no prerequisite is re-derived here.

Editorial note. Isolated visual overlay: only the two original six-strand braids inside the one source center are rendered from retained author-PDF page449 vector clips. The original pinned statement/proof tokens and literal equality remain unchanged; the original braids syntax is not executed. The whole author PDF build-to-commit relationship is unestablished; the root explicitly accepted this locally matched figure as an asset only. This sample is not READY or publication approval.

Allowed standard prerequisites (editorial).

- Prerequisites named by the author inside the counted span. The counted span is reproduced verbatim, including the author's own cross-reference commands. Its prerequisites are the earlier results of the same pinned commit that the author cites there; they are pointed at, never re-proved.

Source reference. Original definition 7.1.2: sources/chapter7.tex, source lines 103–103. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. It is the author's own earlier result in the same pinned commit, registered as a non-load-bearing navigation stub. No bibliography citation is involved.

命题 7.5.6. 设 A 是辫幺半范畴 $\mathcal{V}$ 上的双代数. 以上定义使左 A-模范畴 A-Mod 成为幺半范畴, 以 $(\mathbf{1}, \mu_{\mathbf{1}})$ 为幺元. 对于右 A-模范畴 Mod-A 亦复如是.

对偶地, 应用 A 的乘法和幺元可以使左 A-余模范畴 A-Comod 和右 A-余模范畴 Comod-A 成为幺半范畴.

无论对上述哪一种范畴, 映向 $\mathcal{V}$ 的忘却函子总是幺半函子.

Proof. 基于对偶性, 证前半部即可. 检验繁而不难, 略述如下. 首务是验证纯量乘法符合定义 7.1.2 的交换图表. 对于 $(\mathbf{1}, \mu_{\mathbf{1}})$, 这些性质归结为 $\epsilon : A \to \mathbf{1}$ 是代数的态射. 对于 $M_1 \otimes M_2$ 的模结构, 关于以幺元作纯量乘法的交换图表归结为 $\Delta : A \to A \otimes A$ 是代数的态射, 特别地, 它和 η, $\eta \otimes \eta$ 兼容. 对纯量乘法结合律稍加思索, 可见问题在于验证从 $A \otimes A \otimes M_1 \otimes M_2$ 按两种方式映至 $M_1 \otimes M_2$ 是相等的:

(a) 先作 $\Delta \otimes \Delta \otimes \mathrm{id}_{M_1} \otimes \mathrm{id}_{M_1}$, 再将 $A \otimes A \otimes (A \otimes A \otimes M_1 \otimes M_2)$ 的第一和第三 (或第二和第四) 个 A 依序左乘入 M_1 (或 M_2), 精确到辫结构;

(b) 先对前两个位置作 $\mu : A \otimes A \to A$, 再作 $\mu_{M_1 \otimes M_2}$. 因为

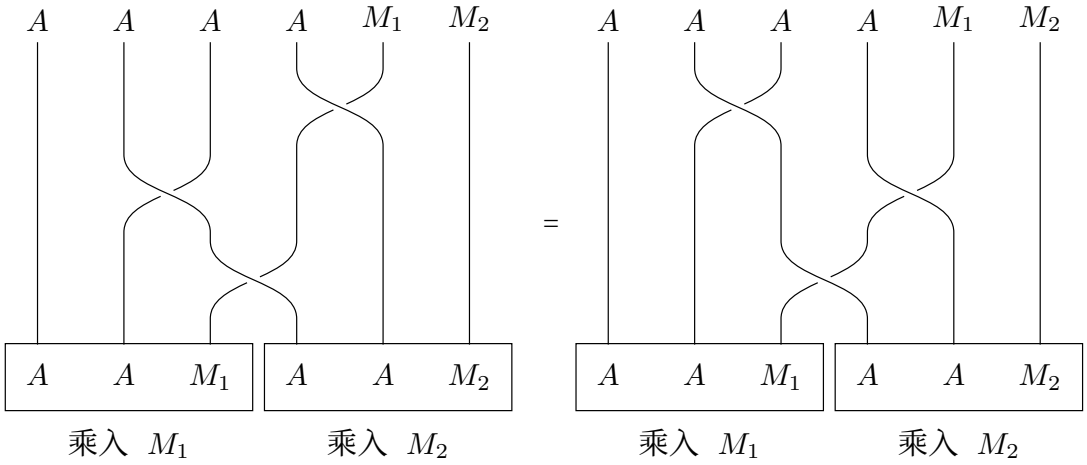
$$\Delta\mu = (\mu \otimes \mu)(\mathrm{id} \otimes c(A, A) \otimes \mathrm{id})(\Delta \otimes \Delta),$$

这等于合成

$$\begin{aligned} A\otimes A\otimes M_1\otimes M_2 &\xrightarrow{\Delta\otimes\Delta\otimes \mathrm{id}_{M_1}\otimes \mathrm{id}_{M_2}} A\otimes A\otimes A\otimes A\otimes M_1\otimes M_2\\ &\xrightarrow{\mathrm{id}_A\otimes c(A,A)\otimes \mathrm{id}_A\otimes \mathrm{id}_{M_1}\otimes \mathrm{id}_{M_2}} \boxed{A\otimes A\otimes A\otimes A}\otimes M_1\otimes M_2\\ &\xrightarrow{\mu\otimes\mu\otimes \mathrm{id}_{M_1}\otimes \mathrm{id}_{M_2}} A\otimes A\otimes M_1\otimes M_2\\ &\xrightarrow{\mathrm{id}_A\otimes c(A,M_1)\otimes \mathrm{id}_{M_2}} A\otimes M_1\otimes A\otimes M_2 \xrightarrow{\mu_1\otimes\mu_2} M_1\otimes M_2. \end{aligned}$$

纯量乘法 μ_1 和 μ_2 各自有结合律而 c 有函子性, 故上式的最后三个箭头也相当于加框部分的第一, 二 (或第三, 四) 个 A 依序左乘入 M_1 (或 M_2), 精确到辫结构.

以辫图帮助理解, 如 [1, §7.4 最后], 证明 (a) = (b) 归结为证[1]



一目了然.

为了说明 A-Mod 为幺半范畴, 尚须证 $\otimes$ 的结合约束和幺元约束都是左 A-模的态射. 前者不难, 后者所需的是余代数的性质 $(\mathrm{id}\otimes\epsilon)\Delta=\mathrm{id}_A=(\epsilon\otimes\mathrm{id})\Delta$, 前提是等同 $A\otimes\mathbf{1}\simeq A\simeq\mathbf{1}\otimes A$. □

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[1] 一点无害的差异: 上引文献中, 态射由下到上作合成, 此处相反.

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